DATA INTEGRATION AND ANALYTICS

Outline:

1. Data Warehousing Concepts

1.1. Data Warehouse, Data Lake, Data Mart
1.2. ETL (Extract, Transform, Load)
1.3. Load-First Integration vs Model-First Integration(diff.)
1.4. Star Schema & Snowflake Schema (with conceptual model drawing)
1.5. Fact Tables & Granularity
1.6. Mini Dimension

2. OLAP (Online Analytical Processing)

2.1. OLAP Basics
2.2. OLAP Operations (Roll-up, Drill-down, Slice, Dice, Pivot)
2.3. OLAP vs OLTP
2.4. Cube Example
2.5. Explanatory OLAP
2.6. Localize View → Globalize View
2.7. Advanced Multidimensional Models

3. Data Modeling & Business Intelligence

3.1. Multidimensional Modeling
3.2. Business Intelligence (BI) Concepts
3.3. Steps in BI
3.4. BI vs Big Data (difference)
3.5. Digital Twin
3.6. FAIR Principles

4. Data Preprocessing & Tools

4.1. Data Preprocessing & Cleaning

4.2. Tools: PDI (Pentaho Data Integration)

4.3. BPMN for ETL Flow

5. Semantic Web & Knowledge Representation

5.1. Knowledge Graph

5.2. RDF (Resource Description Framework) – Details

5.3. Katja er boi theke Chapter 1 → RDF (Complete)

5.4. Dimensions & Hierarchies

5.5. OWL – Why Semantic Data?

5.6. Triple Pattern Form

5.7. Traditional SQL and SPARQL

6. Data Science Process

6.1. Data Science Workflow (from slides)

6.2. Scenario Modeling

6.3. Use Case Scenarios (textual, graph-based data)

7. Linked Data & Vocabulary

7.1. Scenario-Based Modeling

7.2. QB4OLAP Vocabulary (for multidimensional cubes on RDF)

scenario দিলাম → modeling করে দিবে।

" " → ETL করবে। flow আঁকবে

problem

use case →

query দিলাম → H1, ~~SPARQL~~ SPARQL করবে.

Advance → ~~SPARQL~~ OLAP

scenario দিবে → semantic diagram

- [ ] A sample CT Question:

Zilla	Upazilla	Union	village	**Population**	Muslim	Hindu	Christian	Buddhist

Q. Figure 5 shows the demo data to be published in semantic and analytical format. Complete the following task:

1. Sketch a schematic diagram by analyzing the data given in Figure 5
2. Represent your schema using the RDF Model. Use the QB4OLAP vocabulary given in Figure 4 to annotate the schema with multidimensional semantics.
3. Show a sample level member in RDF.
4. Show a sample observation in RDF.